Malihe Aliasgari, Ph.D.

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Research Areas

- Artificial intelligence and machine learning; large language models and recommender systems; information theory, privacy, and distributed computing; coding theory; lattices, Gröbner bases, and algebraic decoding.

Academic Appointments

2023 – Present **Assistant Professor**, Department of Mathematical Sciences, Kean University, Union, NJ.
2021 – 2023 **Postdoctoral Researcher**, School of Integrative Science and Technology, Kean University, Union, NJ.
2020 – 2021 **Adjunct Professor**, School of Integrative Science and Technology, Kean University, Union, NJ.
Jan.–Jul. 2014 **Visiting Researcher**, Carleton University, Ottawa, Canada.
2013 – 2016 **Assistant Professor**, Islamic Azad University, Roudehen Branch, Tehran, Iran.

Education

2016 – 2020 **Ph.D., Electrical Engineering**, New Jersey Institute of Technology (NJIT), GPA: 4.0/4.0.
Dissertation: *Coding Against Stragglers in Distributed Computation Scenarios*.

2010 – 2014 **Ph.D., Pure Mathematics**, Amirkabir University of Technology, Iran, GPA: 4.0/4.0.
Dissertation: *Decoding Lattices via Gröbner Bases*.

2006 – 2008 **M.Sc., Pure Mathematics**, Amirkabir University of Technology, Iran, GPA: 4.0/4.0.
Thesis: *Persistent Homology and Localized Homology*.

2002 – 2006 **B.Sc., Pure Mathematics**, Amirkabir University of Technology, Iran, GPA: 3.8/4.0.

Publications

Manuscripts Under Review / Submitted

- *Encoding-Sensitive Nullstellensatz Certificates for Bounded-Support Syndrome Exclusion in LDPC Codes*, M. Aliasgari, H. Khodaiemehr, and K. Bagheri. Submitted to *IEEE Information Theory Workshop (ITW) 2027*.
- *Reimagining the Newsvendor Game: A Generative AI Approach to Inventory Decision*, S. Shokoohyar, M. Aliasgari, and Y. Nejatbakhsh. Submitted to *Decision Sciences Journal of Innovative Education*, 2026.
- *Query-Formulation Sensitivity in Citizen-Facing Public-Health RAG: A Five-Retriever Retrieval Comparison and a Single-Configuration End-to-End Study*, M. Aliasgari, Y. Nejatbakhsh, and P. Safaee. Under review at *Discover Artificial Intelligence*, September 2026.
- *A Safety-Constrained Hybrid Food Recommender System for Phenylketonuria: Integrating Cuisine-Aware Filtering and Clinical Rule Enforcement*, M. Aliasgari, M. Ajetunmobi, Y. Nejatbakhsh, and R. Nejatbakhsh. Under review at *JMIR Formative Research*, April 2026.
- *Low-Unsatisfaction Screening for QC-LDPC Code Selection in Wireless Links*, M. Aliasgari. Manuscript under revision following review at *IEEE Wireless Communications Letters*, 2026.
- *Functional Converses for Private Matrix-Product Retrieval under Library-Wide Decoder-Side Correlation*, M. Aliasgari and Y. Nejatbakhsh. Submitted to *IEEE Transactions on Information Theory*, August 2026.

Peer-Reviewed Journal Articles

- *Routing Leakage in Adaptive Coded Distributed Matrix Multiplication: An Information-Theoretic Framework for Conditional Differential Privacy*, M. Aliasgari. *IEEE Access*, vol. 14, pp. 100918–100934, 2026.
- *Enhancing Stock Market Prediction with Hybrid Deep Learning: Integrating LSTM, Transformer Attention, Federated Learning, and Sentiment Analysis*, Y. Nejatbakhsh and M. Aliasgari. *IEEE Access*, vol. 14, pp. 3926–3942, 2025.
- *Linear Label Code of a Root Lattice Using Gröbner Bases*, M. Aliasgari, D. Panario, and M.-R. Sadeghi. *Applicable Algebra in Engineering, Communication and Computing*, vol. 35, pp. 3–15, 2024.
- *Private and Secure Distributed Matrix Multiplication with Flexible Communication Load*, M. Aliasgari, O. Simeone, and J. Kliewer. *IEEE Transactions on Information Forensics and Security*, vol. 15, pp. 2722–2734, 2020.
- *Coded Computation Against Processing Delays for Virtualized Cloud-Based Channel Decoding*, M. Aliasgari, J. Kliewer, and O. Simeone. *IEEE Transactions on Communications*, vol. 67, pp. 28–38, 2019.

Publications (continued)

- *An Algebraic Method for Decoding q-ary Codes via Submodules of $\mathbb{Z}^n$*, M. Aliasgari and M.-R. Sadeghi. *IEEE Communications Letters*, vol. 18, no. 5, pp. 857–860, 2014.
- *Gröbner Bases for Lattices and an Algebraic Decoding Algorithm*, M. Aliasgari, M.-R. Sadeghi, and D. Panario. *IEEE Transactions on Communications*, vol. 61, pp. 1222–1230, 2013.

Peer-Reviewed Conference Proceedings

- *The Voice That Tells in Different Languages: A Structured Review of Speech Biomarkers for Cognitive Decline in Ageing, MCI, and Dementia*, P. Safaee, K. Humlen, M. Aliasgari, Y. Nejatbakhsh, M. Mohammadi, L. Beijer, M. Gholami, and M. Noroozian. *Living Adaptations: Housing and Care Innovation*, Antwerp, Belgium, May 2026.
- *Recommending Healthy Food for Phenylketonuria with Collaborative and Content-Based Filtering*, M. Aliasgari, H. Li, Y. Zhang, Y. Nejatbakhsh, and R. Nejatbakhsh. *IEEE International Conference on E-health Networking, Application & Services (HealthCom)*, pp. 546–552, Japan, 2024.
- *Distributed Coded Computation*, M. Aliasgari. 25th Conference on Applications of Computer Algebra (ACA), Gebze–Istanbul, Turkey, 2022.
- *Linear Label Code of a Lattice Using Gröbner Bases*, M. Aliasgari, D. Panario, and M.-R. Sadeghi. 25th Conference on Applications of Computer Algebra (ACA), Gebze–Istanbul, Turkey, 2022.
- *Distributed and Private Coded Matrix Computation with Flexible Communication Load*, M. Aliasgari, O. Simeone, and J. Kliewer. *IEEE International Symposium on Information Theory (ISIT)*, pp. 1092–1096, 2019.
- *Coded Computation Against Straggling Decoders for Network Function Virtualization*, M. Aliasgari, J. Kliewer, and O. Simeone. *IEEE International Symposium on Information Theory (ISIT)*, pp. 711–715, 2018.
- *Johnson-Type Bound for Group Codes and Lattices*, M. Aliasgari, M.-R. Sadeghi, and D. Panario. Conference on Applications of Computer Algebra (ACA), 2015.
- *Binomial Ideal Associated to a Lattice and Its Label Code*, M. Aliasgari, M.-R. Sadeghi, and D. Panario. Conference on Applications of Computer Algebra (ACA), 2015.
- *Minimizing Exceptional Elements and Voids in a Cell Formation Problem with Optimal Number of Cells*, M. Aliasgari, M. Kazemi, R. Kia, and A. Ebrahimi. International Conference on Computers and Industrial Engineering (CIE45), Metz, France, pp. 1286–1294, 2015.
- *Gröbner Bases for Lattices and an Algebraic Decoding Algorithm*, M. Aliasgari, M.-R. Sadeghi, and D. Panario. Forty-Ninth Annual Allerton Conference on Communication, Control, and Computing, Illinois, USA, pp. 1414–1415, 2011.

Selected Research Presentations and Posters

2025
- *PKU-Recipe AI: Personalized PKU-Safe Recipe Generator*, IUSE/IST Research Day, poster presentation.
- *A PKU-Aware Hybrid Recommender System for Safe and Personalized Dietary Management*, IST Research Day, poster presentation.
- *A Smart Assistant for Phenylketonuria (PKU)*, IST Research Day, poster presentation.
- *PKU Food Checker: A Telegram-Based AI Assistant for PKU Dietary Monitoring*, IST Research Day, poster presentation.
- *PKU Scanner AI: A Mobile Tool for Safe Food Scanning*, IUSE/IST Research Day, poster presentation.
- *Using n8n to Create a Phenylketonuria AI Assistant*, IST Research Day, poster presentation.
- *A Hybrid Recommender System for Personalized Dietary Management in Phenylketonuria (PKU)*, Kean Research Day, poster presentation.
- *LLM-Powered Personalized Dietary Agent for Phenylketonuria*, Kean Research Day, poster presentation.
- *Virtual Avatar Development System*, Kean Research Day, poster presentation.
- *Automated Dementia Behavior Analysis and Empathetic Mapping Using Large Language Models*, Kean Research Day, poster presentation.

2024
- *LLM-Powered Personalized Dietary Agent for Phenylketonuria*, IUSE Scholars Academy, Kean University, poster presentation.
- *Linear Label Code of a Lattice*, Third Annual Kean University End-of-Year Faculty Research Conference.

2023
- *Stock Forecasting Using Advanced LSTM Models*, IST Research Day, Kean University, poster presentation.
- *The Power of Artificial Intelligence for Phenylketonuria*, IST Research Day, Kean University, poster presentation.
- *Comparing Predictive Models for Stock Market Analysis: Evaluating the Excellence of LSTM*, Kean Research Day, poster presentation.
- *Private Distributed Computation*, MAA New Jersey Section, Kean University, NJ.

2022
- *Private Distributed Coded Computation*, Applications of Computer Algebra (ACA), Turkey.
- *Linear Label Code of a Root Lattice Using Gröbner Bases*, Applications of Computer Algebra (ACA), Turkey.

2019
- *Distributed and Private Coded Matrix Computation with Flexible Communication Load*, North-American School of Information Theory, Boston, MA, poster presentation.

Publications (continued)

- *Distributed and Private Coded Matrix Computation with Flexible Communication Load*, IEEE ISIT, Paris, France.

2018
- *Coded Computation Against Processing Delays*, Center for Wireless Information Processing Graduate Student Seminar, NJIT.

2017
- *Coded Computation for Distributed Channel Decoding*, Center for Wireless Information Processing Graduate Student Seminar, NJIT.
- *Coded Computation for Distributed Channel Decoding*, North-American School of Information Theory, Atlanta, GA, poster presentation.

2016
- *Gröbner Bases for Lattices and Algebraic Decoding Algorithm*, Graduate Student Seminar, NJIT.

2013
- *On the Ideal Associated to a Lattice and Gröbner Bases of Some Root Lattices*, Groups, Group Rings, and Related Topics, United Arab Emirates University.
- *Maximum Likelihood Decoding for q-ary Codes via Submodules of $\mathbb{Z}_n$*, 44th Annual Iranian Mathematics Conference, Mashhad, Iran.
- *On the Binomial Ideal Associated to a Lattice and Its Label Code*, 44th Annual Iranian Mathematics Conference, Mashhad, Iran.

2012
- *An Algebraic Method for Solving the Closest Vector Problem in a Lattice*, 22nd Iranian Algebra Seminar, Hakim Sabzevari University, Iran, poster presentation.

2010
- *Decoding Group Codes with Gröbner Bases*, 42nd Annual Iranian Mathematics Conference, Rafsanjan, Iran, poster presentation.

Patent

2020
- **M. Aliasgari**, O. Simeone, and J. Kliewer. *Method for Private and Secure Distributed Matrix Multiplication*. United States Patent, U.S. Patent ID D20-043, issued April 28, 2020.

Grant Activity

External Proposals

2026
- *Advancing Early Childhood PKU Care Through Parenting Education and AI*. Caplan Foundation for Early Childhood. **M. Aliasgari**, Y. Nejatbakhsh, and R. Singh. Amount requested: $50,000. Submitted January 2026.

2025
- *PRECISE: Personalized AI Recommendations for Empowering Care in Supporting Everyday Life Diet and Behavior in Adults with PKU*. NPKUA. **M. Aliasgari**, Y. Nejatbakhsh, and R. Singh. Originally submitted June 2024; not funded; resubmitted June 2025.

Proposal Consulting

- Consultant, NSF proposal *Length Scale Dependence of Extracellular Matrix Alignment in Tumor Growth and Invasion*. PI: Amir K. Miri, New Jersey Institute of Technology. Proposal amount: $600,000. Submitted 2025.

Awards, Fellowships, and Research Support

2026
- June Activities Award, Kean University, $2,000.
- Graduate Research Assistantship support, Kean University.

2025
- IST Fellowship, Kean University, $10,000.
- Participant, Kean University Shark Tank, *Smart Mind Care*.

2024
- Participant, Kean University Shark Tank, *AI for PKU*.
- Nominee, Faculty Research Mentor of the Year Award (Early Career Faculty), Kean University.

2023
- IST Fellowship, Kean University, $10,000.

2019
- Summer Research Assistantship Award, NJIT, $5,000.

2018–2019
- Research Assistantship Award, NJIT, $32,000.

2018
- Summer Research Assistantship Award, NJIT, $5,000.

2017–2018
- Research Assistantship Award, NJIT, $33,000.

2017
- Travel Award, NJIT, $600.

2016–2017
- Research Assistantship Award, NJIT, $52,000.

2015
- CIE45 Travel Award, Islamic Azad University, Roudehen Branch, $2,000.

2014
- NSERC Grant support, Carleton University, $12,000.

Awards, Fellowships, and Research Support (continued)

2013 Sabbatical Award, Amirkabir University of Technology, $10,000.

Teaching Experience

Kean University

Fall 2026 Precalculus; Calculus I.

Summer 2026 Large Language Models.

2025 Detection and Estimation Theory: Inference and Machine Learning for Engineers; Introduction to Deep Learning.

Spring 2024 Deep Learning and Recommender Systems.

2024–2025 Discrete Mathematics.

2023–2024 Research Federate Initiative (RFI).

2022–2023 Special Topics: Machine Learning.

2022 Information Theory and Coding.

2020–2024 Mathematics/Computational Methods I; Precalculus; Mathematics/Computational Methods II; Calculus I; associated laboratories.

New Jersey Institute of Technology

2019 Digital Communication Systems.

Earlier Teaching

2014–2015 LaTeX and Beamer Workshops for faculty, Islamic Azad University, Tehran, Iran.

2013 Engineering Mathematics and Ordinary Differential Equations, Islamic Azad University, Iran; Error Control Coding, Amirkabir University of Technology.

2009–2012 Ordinary Differential Equations and Calculus II, Amirkabir University of Technology / Shahid Rajaee Teacher Training University / Islamic Azad University, Iran.

2009 Statistics and Calculus I, Islamic Azad University, Iran.

2006–2008 Mathematics Olympiad instruction, Iran.

Teaching Assistantships

2018 Stochastic Signals and Systems, New Jersey Institute of Technology.

2017, 2019 Communications Systems I (M.S.), New Jersey Institute of Technology.

2009 Advanced Algebra (M.Sc.), Shahid Rajaee Teacher Training University, Tehran, Iran.

Research Mentoring

2026 Group Summer Scholars Research Program (GSSRP), high-school student research mentoring, Kean University.

UR2PhD program mentoring, Kean University.

Graduate research assistant mentoring, Kean University.

2024–2026 IUSE Scholars Academy student research mentoring, Kean University.

2025 Group Summer Scholars Research Program (GSSRP), high-school student research mentoring, Kean University.

UR2PhD program mentoring, Kean University.

2023–Present Master's thesis mentoring.

2022–Present Undergraduate student research mentoring.

Professional Service

University and Community Service

2024–Present Computational Science Ph.D. Curriculum Committee.

2023 Judge, BRHS Hackathon.

2023–Present Research Committee, WKU.

Editorial Boards

2025–Present Editorial Board, *Mathematics and Computer Science*.

2019–Present Editorial Board, *Electrical Engineering and Computer Science (EECS)*.

Professional Service (continued)

Peer Review

- Reviewer for *IEEE Transactions on Neural Networks and Learning Systems*; *Journal of Economic and Social Measurement*; *IEEE Access*; *IEEE Journal on Selected Areas in Information Theory*; *IEEE International Symposium on Information Theory*; *IEEE Transactions on Information Forensics and Security*; *IEEE Transactions on Communications*; *IEEE Transactions on Information Theory*; *Designs, Codes and Cryptography*; IEEE SPAWC; IEEE WCNC; CECNet; and ISCISC.

Professional Memberships

- Institute of Electrical and Electronics Engineers (IEEE); IEEE Information Theory Society.

Technical Skills

Programming / Computing	Python, MATLAB, CoCoA, Magma.
Mathematical / Technical Tools	Mathematica, Maple, Vensim, LaTeX, Inkscape.
Languages	English, Persian, Azerbaijani, Spanish.